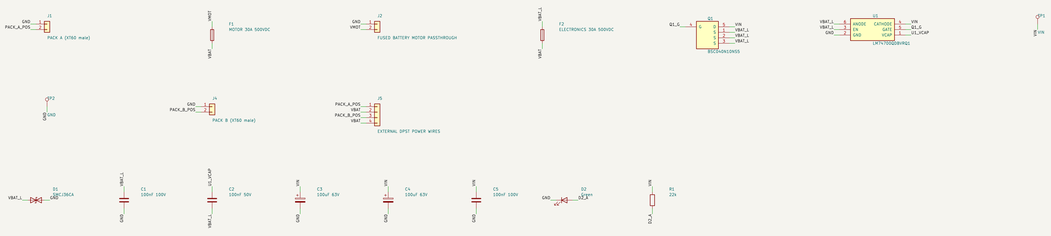
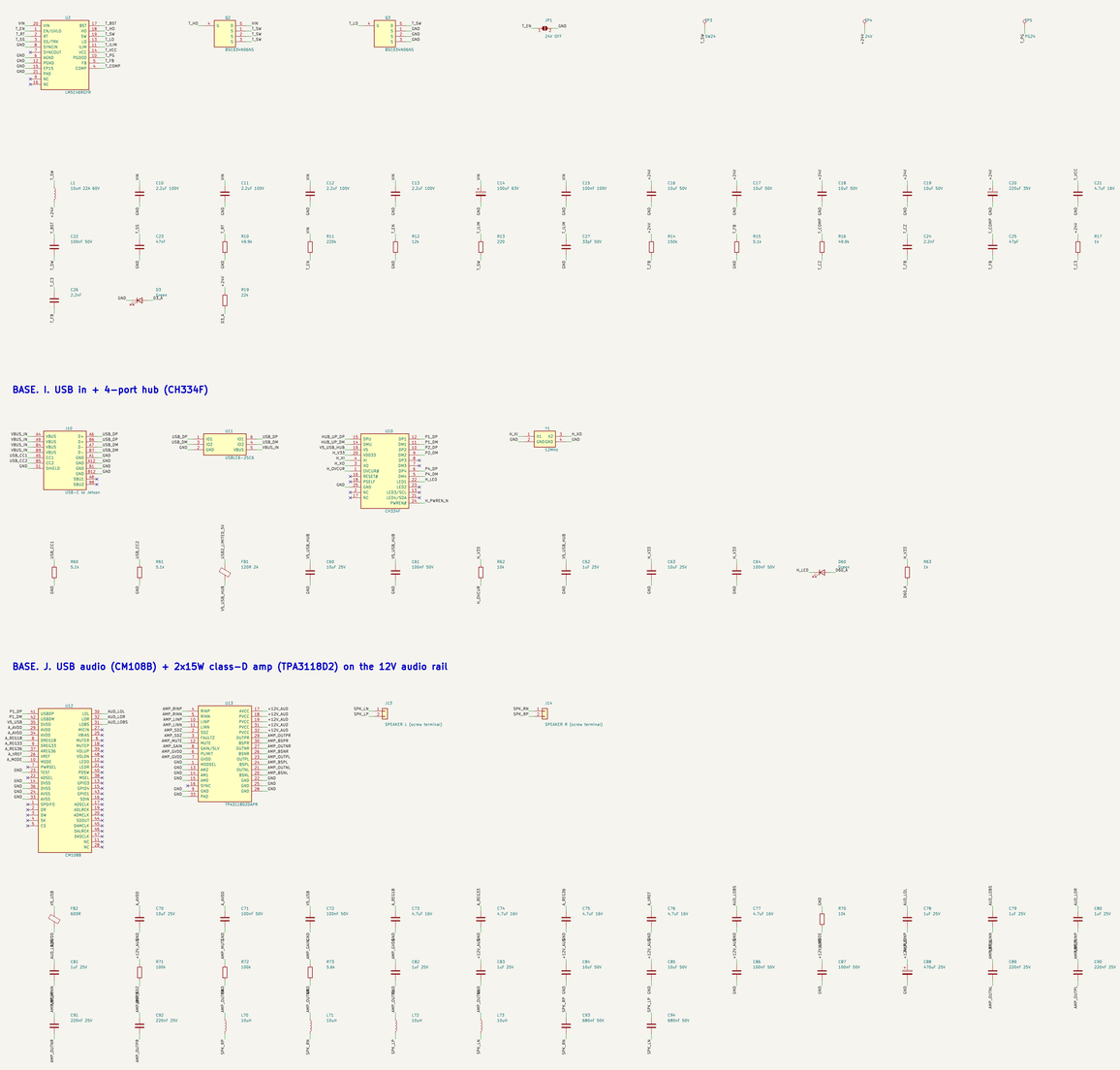
BASE. A. Battery input, motor pass-through, logic protection



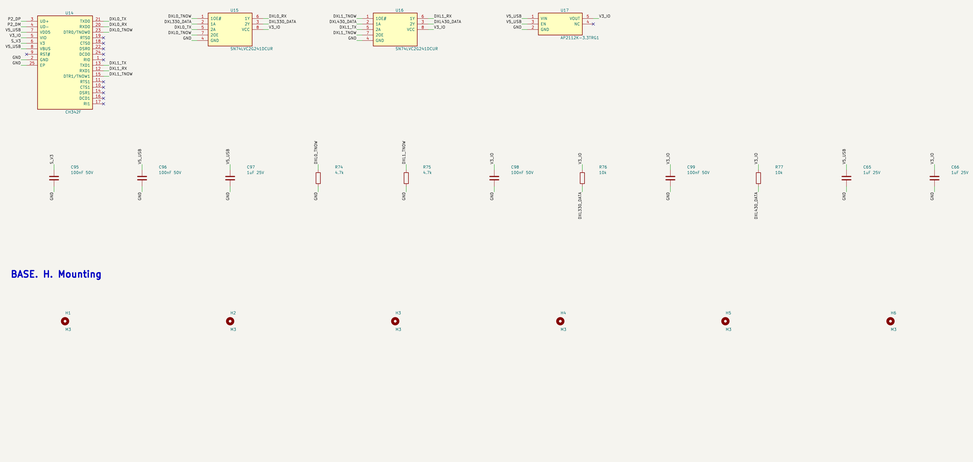
BASE. B. 24V for Thor dev kit (8A connector ceiling; unqualified) (LM5146 sync buck, 200 kHz)



BASE. I. USB in + 4-port hub (CH334F)

BASE. J. USB audio (CM108B) + 2x15W class-D amp (TPA3118D2) on the 12V audio rail

BASE. K. Dynamixel USB interface: CH342F dual serial + half-duplex buffers (U2D2 replacement)



BASE. H. Mounting

IO. A. Input bulk



IO. A. Servo pulse bulk

IO. D. Three5V PWM outputs, default disabled

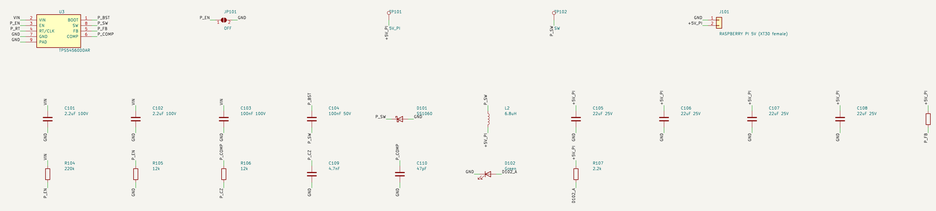


IO. E. Service

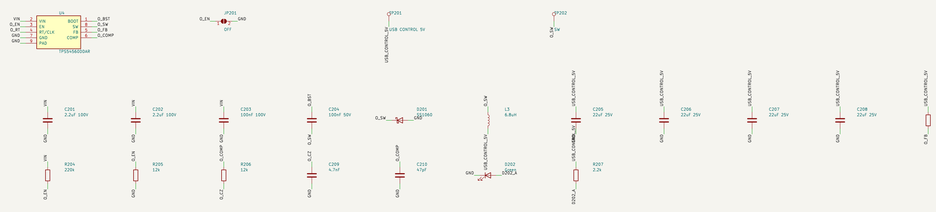
USB2. USB2 CONTROL. Local supply and host-controlled data attachment

SERVICE. Optional local regulator OFF shunts

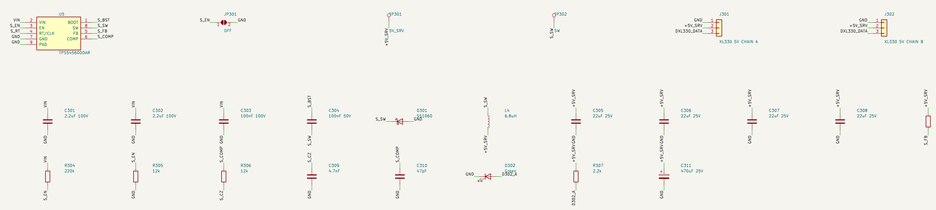
BASE. C. 5V / 5A Raspberry Pi



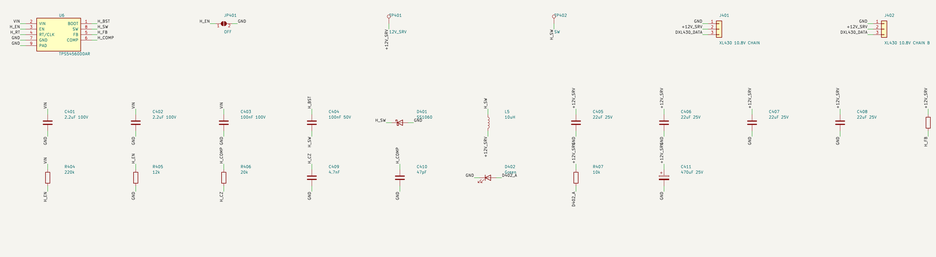
BASE. D. 5V / 5A OAK-D depth camera



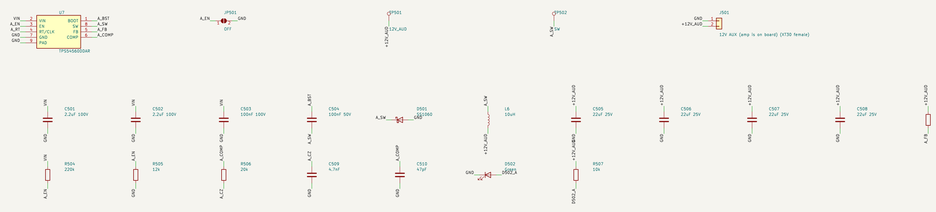
BASE. E. XL330 servo power, 5 V, two daisy-chain sockets



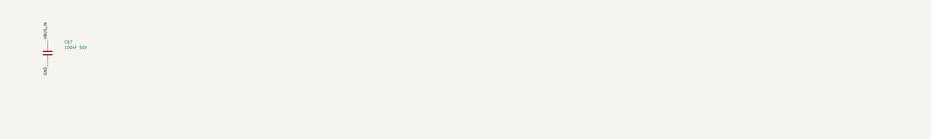
BASE. F. XL430 servo power, 10.8 V, one daisy-chain socket



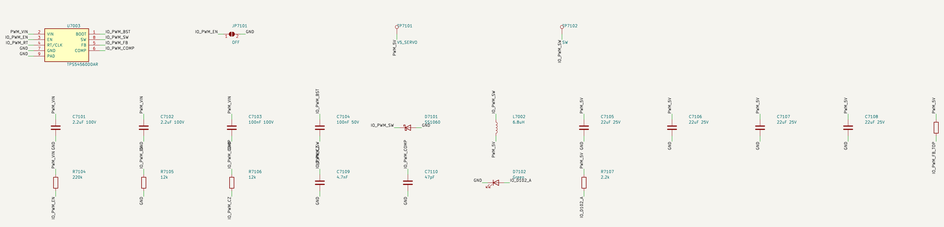
BASE. G. 11.7V / 5A Audio amplifier



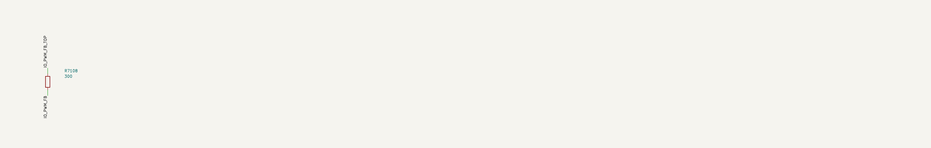
BASE. I. USB attachment slew control



IO. A. Dedicated MG90S servo5V (5A target unqualified)

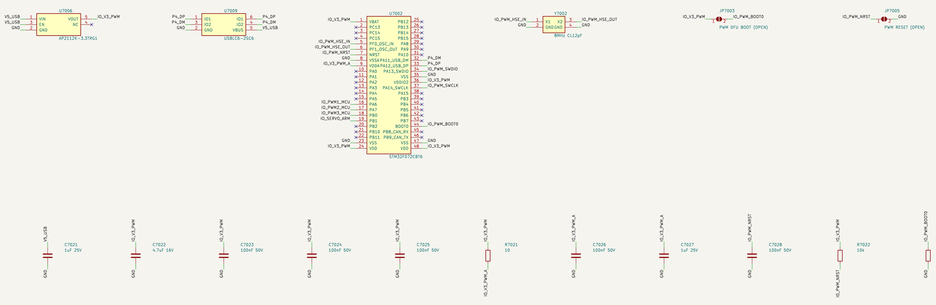


IO. A. Servo5V trim to4.992V nominal



IO. A. Local input surge footprint

IO. B2. PWM USB MCU and independent USB logic supply



IO. E. Mounting

USB2. USB2 CONTROL. Local supply and host-controlled data attachment

BRANCH. PWM converter input fuse

SERVICE. Optional local regulator OFF shunts

THOR OUTPUT. Standard four-contact power cable

MOUNT. Harness support

ENGINEERING PROTOTYPE: final layout/assembly review and bench qualification required. External mechanical DPST disconnect; STOP ROBOT BEFORE OFF. 27-33V intended battery operation,33.6V full; individual pack-source fuses required. J2 is original fused BATTERY-VOLTAGE motor passthrough to existing Flipsky controllers. No motor converter, brake clamp, external brake resistor, thermal contact or onboard CAN. Existing motor controllers retain their previous control connection and braking configuration. USB2 VBUS is sense only; local5V powers audio/UART/PWM through the CH334 hub. Hub port3 unused; no configuration EEPROM. OAK-D/microphone connect externally to Jetson USB3. Seven local shunts shortENtoGND=OFF; they do not i